

Constrained Monocular Vision-Guided Handshake Interaction for Quadruped Robots

Liangbin Wu^{1,†}, Shouchen Chen^{2,†}, Yihuang Zheng², Hongxin Chen²
and Xiaowei Chen^{3,*}

¹ School of Information Engineering, Fujian Polytechnic of Water Conservancy and Electric Power, Fujian, China

² College of Computer and Cyber Security, Fujian Normal University, Fujian, China

³ Fujian Provincial Key Laboratory of Network Security and Cryptology, College of Computer and Cyber Security, Fujian Normal University, Fujian, China

E-mail: chenxw@fjnu.edu.cn

Abstract. Front-paw human-robot interaction depends on a continuous chain from visual perception to paw control and recovery. We built a constrained interaction system for a quadruped robot using only its built-in front monocular RGB camera. Hand landmarks define a two-dimensional image target inside an operator-guided interaction region, which drives a 50 Hz paw policy after filtering. An approximately 500 Hz low-level loop enforces joint and safety limits. In ten consecutive prone trials, a non-operator observer rated contact and contact lasting at least 0.6 s in 10/10 trials. Logs recorded 9/10 clean execution completions and 10/10 safe withdrawals. In Trial 2, the paw contacted the hand before the tracking-error gate triggered at 0.6004 rad; the robot then withdrew safely. These trials show that the system can operate under the tested constraints. They do not establish unrestricted three-dimensional reaching, cross-scene generalization, or stability improvement. Code and structured research materials are available at <https://github.com/tuoLuoSuan/go2-visual-paw-interaction>.

1. Introduction

Quadruped legs have been used for learned motor control, foot tracking, and visually guided manipulation [10, 5, 6, 7]. Quadruped handshaking motions have also been studied [4], and hand gestures have teleoperated an arm-equipped quadruped [9]. We ask a narrower question: can a quadruped robot connect its built-in monocular vision to active front-paw interaction while keeping visible contact, control completion, and safe withdrawal auditable and separate?

MediaPipe Hands estimates 21 landmarks from monocular RGB images [1], but these image features do not directly provide metric depth. Visual-servo research distinguishes image-feature feedback from control based on estimated pose [2, 3]. We use the former only within a predefined indoor region: normalized image coordinates modify the paw target, but are not interpreted as a general robot-frame position.

Physical interaction also needs distinct evidence channels. Repeated torque triggers during one contact are not independent observations, and visible contact does not show whether the program completed. In line with the separation of collision-detection functions described by

† Liangbin Wu and Shouchen Chen contributed equally to this work and share first authorship.

* Corresponding author.

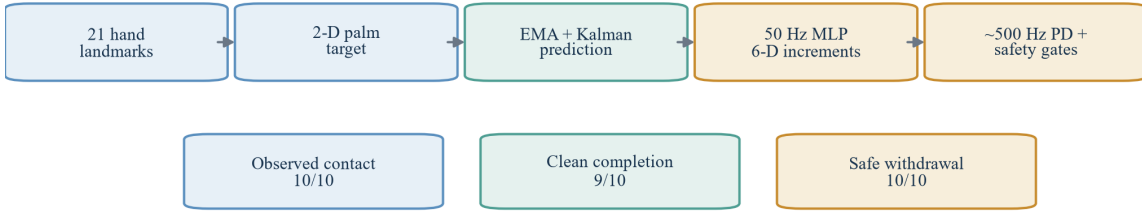
Haddadin et al. [11], video rates contact, logs determine clean completion, and recovery records determine safe withdrawal. Our implementation connects monocular perception, target conditioning, a learned paw policy, low-level control, and fail-closed recovery in one auditable pipeline. The standing demonstration appears separately in Figure 1 and is excluded from the prone denominator.



(a) Prone main experiment.



(b) Standing qualitative extension (3/3 recorded trials).



(c) Constrained perception-to-control chain.

Figure 1. System overview. Prone interaction is the main ten-trial experiment; standing is reported separately and excluded from the prone denominator. The control chain keeps observed contact, clean completion, and safe withdrawal as separate endpoints. EMA denotes exponential moving average, and PD denotes proportional-derivative control.

2. System and method

2.1. Task scope and perception

The robot began prone with both front paws forward. The operator presented one hand approximately 30–50 cm from the robot and 10–25 cm above the floor. These distances describe the setup, not program estimates. The side-only route used normalized image coordinates without camera-to-body extrinsics. Approximately 3 s of target loss, timeout, or a safety-gate event initiated withdrawal and released low-level control.

The perception process acquired JPEG samples through the Unitree video interface, decoded and resized each frame to 960×540 pixels, and applied MediaPipe Hand Landmarker. At most two detected hands were retained, with detection, presence, and tracking thresholds of 0.5. If two hands were present, the larger image-space hand was selected. For the palm landmark set $\mathcal{P} = \{0, 5, 9, 13, 17\}$, the target was

$$(p_x, p_y) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} (x_i, y_i), \quad |\mathcal{P}| = 5, \quad (1)$$

where x_i and y_i are normalized image coordinates. A valid detection produced (p_x, p_y) ; otherwise the process sent a no-target message. An exponential moving average with coefficient 0.25 reduced frame-to-frame variation. A constant-velocity Kalman filter then predicted the two-dimensional target 0.10 s ahead and continued prediction for at most 0.25 s during a short loss.

The image target retains direction within the camera view, not metric hand position. A hand at the same image location can lie at different depths; the operator therefore supplies the distance constraint by presenting the hand in the interaction region. Neither the pixel coordinates nor the side indicator resolve that ambiguity. This distinction also applies to the simulation distances reported below: those distances use simulated geometry, not a depth measurement from the real camera.

For successive valid received targets $\mathbf{p}_k$, smoothing used

$$\bar{\mathbf{p}}_k = (1 - \alpha)\bar{\mathbf{p}}_{k-1} + \alpha\mathbf{p}_k, \quad \alpha = 0.25. \quad (2)$$

The first valid target initialized the smoother. The predictor maintained position and velocity in each image direction, $[p_x, v_x, p_y, v_y]^T$, and used the smoothed coordinates as measurements. Its configured look-ahead was 0.10 s, not an experimentally measured end-to-end latency. An explicit no-target message cleared the stream target and smoother; a lack of new packets for more than 1 s also cleared them. The predictor could bridge a short unavailable-target interval, whereas the longer target-loss gate initiated withdrawal. These separate time scales prevent brief detection loss and a sustained missing target from being treated as the same event; their individual benefit was not tested by ablation.

2.2. Policy and control

Separate perception and control processes exchanged a local User Datagram Protocol message. At 50 Hz, the deployed multilayer perceptron (MLP) received 12 normalized joint angles, 12 normalized joint velocities, p_x , p_y , and an image-side indicator. Its $27 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 6$ architecture used hyperbolic-tangent hidden activations; the six outputs were three-joint increments for both front legs. Only this checkpoint was deployed in the prone trials.

An auxiliary gated recurrent unit (GRU) checkpoint used the same input and output definitions with architecture $27 \rightarrow \text{GRU}(128) \rightarrow 128 \rightarrow 128 \rightarrow 6$. Its hidden state propagated within each evaluation trajectory. The MLP continued from earlier weights, whereas the GRU began from random parameters; with a single training seed per architecture, their comparison cannot isolate architecture.

More explicitly, the deployed observation was

$$\mathbf{o}_t = [\mathbf{q}_t/\pi, \dot{\mathbf{q}}_t/10, p_x, p_y, \mathbf{1}\{p_x < 0.5\}] \in \mathbb{R}^{27}, \quad (3)$$

where the twelve joint angles are in radians and velocities in radians per second before normalization. The final scalar encodes image side; it is not an independently verified paw-selection label. The six policy outputs are ordered as the right-front and left-front hip, thigh and calf increments. Inference uses the deterministic output head, with NumPy evaluating the exported weights. The MLP has no recurrent hidden state, although the perception filters still retain temporal state.

At each policy update, each front-leg target first received its policy increment and was clipped to the configured joint range. A second limiter restricted its change to 0.03 rad relative to the previous target. The low-level loop reused this target between policy updates. Thus, the network does not directly command an unrestricted joint position, Cartesian foot position or contact force. Joint-range limits, incremental motion limits and the tracking-error gate constrain different parts of this chain.

The final stage used 32 parallel environments, 2500 iterations, 200-step rollouts, and 300 imitation warm-up iterations before proximal policy optimization. Training sampled 0.03–0.25 s perception delay and 2.0 N m torque noise. The reward combined target distance, action smoothness, energy, inactive-leg deviation, and active-leg joint error; full optimization settings are released with the code.

Table 1 gives the archived training settings for both checkpoints. Generalized advantage estimation used discount γ and trace parameter λ . The MLP’s earlier checkpoint was not separately preserved, so the final-stage settings document continuation rather than a complete from-scratch training lineage. The GRU training record has no resume checkpoint. Equal final-stage settings therefore do not imply equal total training budgets.

Table 1. Archived final-stage training settings. Both models used training seed 0, but the MLP resumed earlier weights and the GRU started without a resume checkpoint.

Setting	Value for each checkpoint
Parallel environments / iterations	32 / 2500
Rollout length / imitation warm-up	200 steps / 300 iterations
Learning rate	3×10^{-4}
Discount / advantage trace	$\gamma = 0.99$, $\lambda = 0.95$
Perception-delay range	0.03–0.25 s
Torque-noise setting	2.0 N m
Reward scale / smoothness weight	0.03 / 0.15
Active-leg joint-error weight	2.0

An approximately 500 Hz proportional–derivative loop applied joint-range and per-step limits. Reach used $K_p = 120$, $K_d = 10$; tracking used $K_p = 40$, $K_d = 4$, with a 30 N m command-torque limit. The other legs held a recorded prone posture. These were execution constraints, not evidence of improved body stability.

Table 2. Main deployment parameters for the prone real-robot trials.

Item	Formal value
Vision input	Built-in monocular RGB; normalized p_x, p_y
Target processing	EMA 0.25; prediction 0.10 s; short loss 0.25 s
Policy	27 observations; 6 actions; 50 Hz
Low-level loop	Approximately 2 ms (approximately 500 Hz)
Front-leg gains	Reach 120/10; tracking 40/4 for K_p/K_d
Timing	Hold 2 s; initial reach 4 s; tracking at most 90 s
Safety gates	Support deviation 0.25 rad; front-leg error 0.60 rad
Temperature	Start gate 70 °C; hard limit 80 °C

2.3. Contact event and fail-closed recovery

An online event detector monitored estimated torque in the interacting leg. After a 0.5 s warm-up, an event required both 1.0 N m absolute torque and twice the moving-average baseline. Each event set a 0.6 s hold flag. In the archived implementation, this flag gated the hand-mapped update, but did not directly gate the MLP update that preceded it. It therefore cannot establish that the deployed policy stopped moving for 0.6 s. Sustained contact was assessed from video, not inferred from this flag or the detector count.

The action sequence comprised Reach, Track, Retreat, and Restore. Before low-level control, the operator checked the remote controller, space, and temperature. Target loss, the 90 s limit, or a joint, attitude, or temperature gate routed control to Retreat and Restore. Front-leg tracking error was

$$e_t = \max_{j \in \mathcal{J}_{\text{front}}} |q_{j,t}^{\text{cmd}} - q_{j,t}^{\text{meas}}|, \quad (4)$$

and tracking was aborted when $e_t > 0.60$ rad. The attitude gate was 45° absolute roll or pitch. Figure 2 shows the state flow and the trial-level maximum tracking errors.

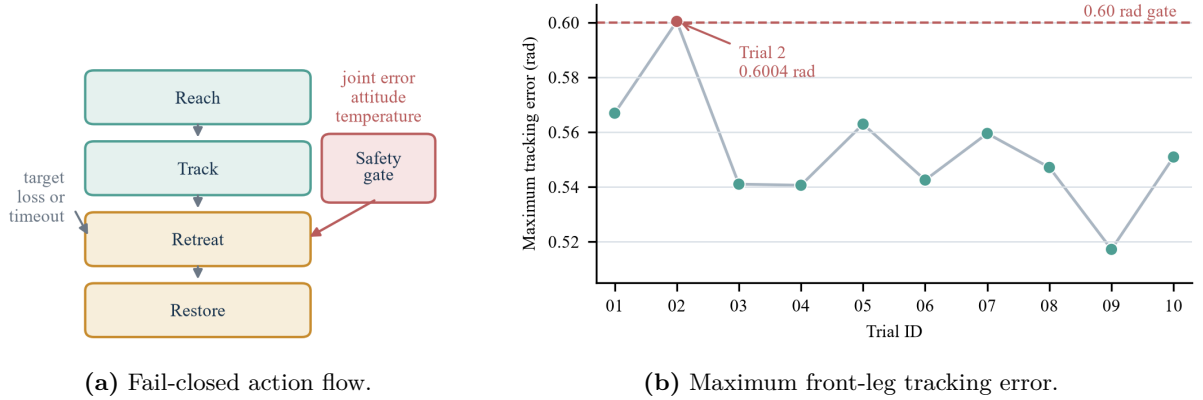


Figure 2. Fail-closed flow and maximum front-leg tracking error. Trial 2 crossed the 0.60 rad gate at 0.6004 rad. The controller then withdrew the paw and restored motion service.

3. Experimental protocol

Experiments used a Unitree GO2 EDU quadruped robot and its built-in monocular camera, with perception and control in an Ubuntu 22.04 virtual machine. During one indoor wooden-floor session, one author operated ten consecutive prone trials, five per image side. The operator announced each trial and maintained the hand target during reach and contact. Every commanded trial, including an abort, remained in the denominator.

A tripod video contained ten numbered segments. One non-operator observer, without algorithm or log access, rated visible hand-paw contact and duration of at least 0.6 s. The side view could not establish paw-selection correctness. One commanded trial was the analysis unit; frames, telemetry, control cycles, and detector events were within-trial observations. Repetition within this single session did not provide independent operator or environment replication.

Video determined contact and sustained contact, logs determined clean completion, and recovery records determined safe withdrawal. Counts and proportions were descriptive because the pilot had one operator, one session, and no real-robot comparator. The original trial-2 record incorrectly marked completion; the preserved log and linked correction classify it as aborted. Thus, the trial remains in every relevant denominator without promoting detector events to trials.

The MLP and GRU used the same 27-dimensional observation, six-dimensional action, evaluator, and evaluation seed. Each ran on 20 trajectories of 200 steps with 0.03–0.15 s delay. These were checkpoint evaluations, not independent training runs, so no significance test was used.

Table 3 states how the real-robot outcomes were read. In particular, an available video rating did not overwrite a missing sensor-ground-truth field in the raw JSON. Conversely, a positive contact rating did not override an abort in the execution log. The original records and the trial-2

correction were retained together, so a reader can reproduce both the reported count and the reason for departing from the original completion field.

Table 3. Decision rules for the prone trial-level readout. Every measured endpoint retains all ten commanded attempts in its denominator.

Endpoint	Evidence and decision rule
Visible contact	Non-operator video rating of hand-paw contact; not a torque-trigger label.
Sustained contact	Observer-rated contact lasting at least 0.6 s; no frame-resolved timing claim.
Clean completion	No execution abort in the run log, with the linked correction applied to trial 2.
Safe withdrawal	Recorded retreat completion, successful mode selection, restored motion-service mode and successful restore return code.
Correct paw selection	Not measured: the side-view video could not resolve the required robot-side correspondence.

For the simulation comparison, let $d_{e,t}$ be the recorded Euclidean foot-target distance in trajectory e at step t . The pooled mean and empirical cumulative distribution were

$$\bar{d} = \frac{1}{N} \sum_{e,t} d_{e,t}, \quad \hat{F}(r) = \frac{1}{N} \sum_{e,t} \mathbf{1}\{d_{e,t} \leq r\}, \quad N = 20 \times 200. \quad (5)$$

The 5 cm measure used the strict condition $d_{e,t} < 0.05$ m. A full run meant reaching 200 steps, not completing a physical handshake. Per-trajectory means show variation across the archived evaluations without treating their 4000 correlated time steps as independent replications. We reconstructed the summaries from the step and trajectory files before plotting them.

4. Results

Visible contact and contact lasting at least 0.6 s were rated in 10/10 prone trials. Logs recorded 9/10 clean completions and 10/10 safe withdrawals. Figure 3 separates these endpoints rather than combining them as one success rate.

In trial 2, visible contact preceded a maximum tracking error of 0.6004 rad. The controller stopped tracking, retracted the paw, released low-level commands, and restored motion service. Contact and safe withdrawal were therefore positive, but clean completion was not. Other trials ranged from 0.5172 to 0.5669 rad.

Under the shared evaluator (Table 4), mean foot-target distance was 0.1053 m for the MLP and 0.1187 m for the GRU; both completed all 20 trajectories. Each had a single training seed, and only the MLP continued from earlier weights. The difference therefore describes these checkpoints, not a general architecture ranking.

In a separate standing protocol, the programmed motion reached visible contact in all 3/3 recorded trials. These trials used an older record format without matched observer annotation, so they remain a qualitative standing extension and are not pooled with the prone trials.

Figure 4 expands the checkpoint comparison beyond the pooled mean. The empirical distributions retain all recorded steps, and the trajectory panel retains all twenty means per model. Both checkpoints completed the evaluation horizon, yet only 10.15% and 9.375% of their steps, respectively, were within 5 cm. Completion of the simulated horizon and close tracking therefore capture different behavior. These simulated distances do not quantify real-robot hand localization or contact accuracy.

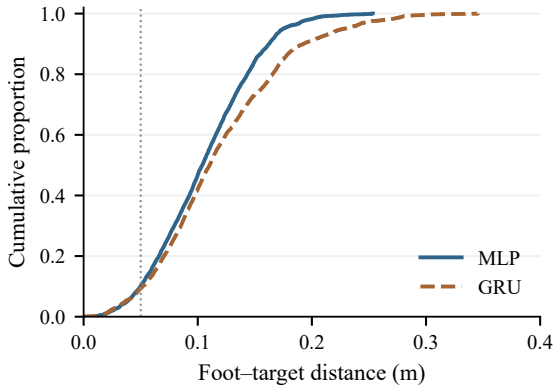
	01	02	03	04	05	06	07	08	09	10
Observed contact	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Hold ≥ 0.6 s	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
Clean completion	OK	ABORT	OK	OK	OK	OK	OK	OK	OK	OK
Safe withdrawal	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK

Trial 2: contact held; execution then aborted

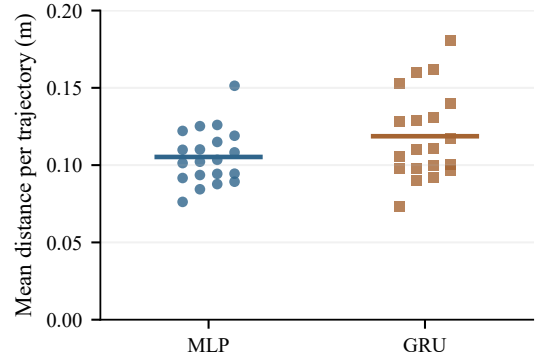
Figure 3. Ten prone trials. Contact was observer rated; completion and withdrawal came from logs. Trial 2 contacted before aborting, and all trials withdrew safely.

Table 4. Descriptive evaluation of two archived policy checkpoints. The percentage is the share of evaluated time steps below 5 cm; full runs count 200-step trajectories. Each checkpoint has one training seed.

Backbone	Mean (m)	Median/P90 (m)	< 5 cm / full runs
MLP	0.1053	0.1039/0.1624	10.15% / 20 of 20
GRU	0.1187	0.1104/0.1927	9.375% / 20 of 20



(a) Pooled step-distance distributions.



(b) All trajectory means.

Figure 4. Descriptive comparison of the two archived checkpoints: 20 trajectories of 200 steps per model. Panel (a) uses all 4000 distances per checkpoint; the vertical reference is 5 cm. In (b), each dot is one trajectory mean and each horizontal segment is the mean across trajectories. Time steps are correlated, and there is only one training seed per architecture, with unequal training origins. No inferential error bars or significance tests are shown.

5. Discussion and limitations

Across the recorded prone trials, the monocular perception-to-recovery chain ran end to end under operator-guided conditions. Image coordinates controlled the paw only inside a restricted region chosen by the operator. This was constrained image-space interaction, not autonomous metric localization.

Trial 2 shows why endpoints must remain separate: contact occurred before the tracking gate aborted execution, followed by safe withdrawal. One success label would hide this sequence, while repeated torque events would falsely inflate the sample count.

The contact-event implementation gives a second reason to keep evidence channels separate. A configured hold interval is not itself evidence of an actual motion pause, particularly when the MLP and hand-mapped branches handle that flag differently. The video rating supports sustained contact in the recorded trials, but does not establish regulated contact force or a stationary paw. Similarly, successful recovery in these ten attempts documents observed withdrawals, not a general safety guarantee.

The checkpoint comparison helps explain the deployed MLP, but it does not rank MLP against GRU in general. Playful DoggyBot studied memory under faster catching, target loss, and delay [8]; our task and endpoints differ. Unequal training origins and one seed each leave initialization, optimization, and checkpoint selection as alternative explanations.

Real-robot evidence is limited to one operator, session, environment, and observer. The side view did not establish paw choice or frame-level contact time. A 70 °C start gate also differed from an earlier 55 °C protocol, partly confounding temperature with trial order. Foot slip, support force, body attitude, and contact force were unmeasured, so the data cannot support stability claims. Testing across operators, sessions, camera views, and environments is still needed. Metric reaching will require depth information or calibrated geometry with an explicit spatial constraint, such as a known interaction plane. A fair architecture comparison will require matched training and multiple seeds.

6. Conclusion

We evaluated a constrained monocular paw-interaction pipeline on the quadruped robot. Across ten prone trials, sustained contact was rated in 10/10, clean completion in 9/10, and safe withdrawal in 10/10; one post-contact abort occurred at 0.6004 rad. The claim is therefore limited to operator-guided two-dimensional image interaction. These data do not establish unrestricted spatial interaction, stability improvement, or a general MLP–GRU ranking.

Code and data availability

Cleaned code, checkpoints, structured records, annotations, figure data, and audit tools are available at <https://github.com/tuoLuoSuan/go2-visual-paw-interaction>. Raw videos, participant material, private settings, the third-party hand model, and camera calibration are excluded. The repository has no archival DOI or open-source licence and is not a complete public dataset.

Acknowledgments

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